numeric system was incomplete. Historians place the concise development of this humble digit on Brahmagupta.

Brahmagupta forged the fundamental rules of the use of zero within mathematical calculations, which was significant because until that time it had merely been used as a placeholder within larger numbers. According to these rules, 1+0=1; 1-0=1; and 1x0=0. Now, large calculations could be carried out easily, be they addition, subtraction, or multiplication. Yet, one of the rules of zero evaded Brahmagupta: that of its division. This puzzle would not be completed for several more centuries, but the works of India's most renowned mathematician had already changed the face of the pursuit of everything from simple pastoral accounting to complex trigonometry and algebra needed for astronomy.

Working further with zeros, Indian mathematicians, Brahmagupta included, discovered a completely hypothetical way to use digits: negative numbers. It was a whole new perspective on math, in that numbers could be used to represent potential values. The first use of negative numbers was to represent debts on accounting sheets. For example, if a customer ordered and paid for six swatches of silk from a merchant, then left with only four, the current value of swatch numbers sits at -2 in favor of the customer. If he or she takes four and pays for 2, it's -2 in favor of the merchant. Negative numbers and their wider use in mathematics wouldn't be fully examined in Europe for another 1,000 years, but Brahmagupta worked with them extensively.

He also worked with positive numbers—specifically, distances and physical measurements—to further develop the Greeks' written language of trigonometry. He published several works in which he criticized his fellow astronomers for using faulty methodology and lay the groundwork for various types of mathematics, including algebra, geometry, and trigonometry. Within the latter field, he described a formula for working out the area of a cyclic quadrilateral—that is a four-sided figure with all four vertices (corners) on the circumference of the circle. Brahmagupta's formula is written as follows, where K is the area of the quadrilateral and a, b, c, and d are lengths of the figure's sides.

$K = ¼ \sqrt{(-a+b+c+d)(a-b+c+d)(a+b-c+d)(a+b+c-d)}$

Brahmagupta and Aryabhata's work was founded on the mathematical calculations of the early Greeks, but they managed more precise